

Businesses stored in the yelp dataset may be classified as “popular” or “successful”. Due to the subjectivity of these terms, the process for identifying businesses belonging to each group must be consistent and reliant upon objective data.

The following definitions will be used for each category:

Popular businesses are those that see the most customers. Popularity will be based on the frequency of check-ins and review postings corresponding to each business id in the database. The more people that interact with a business, whether that be through visits (check-ins) or reviews, the more popular a business is. Popularity does not indicate a high rating, only high interaction. The quality of a business falls under the success qualification instead.

Successful businesses have both high popularity and high satisfaction by customers. As opposed to analyzing only the frequency of visits and reviews, successful business will require high review scores as a distinguishing metric. A business may have high popularity but low review scores, indicating the potential for lower customer loyalty and a worse quality of service. Such qualities distinguish if a business is successful or not.

For each category, businesses must be compared with those in the same category and same zip code area to reduce variables like population size or business type from skewing results. For example, businesses in a large city may have more check-ins and interaction only because they exist in a more populated area, not because the business itself is popular. Similarly, some categories may see more business (like fast-food or a gas station for example) without being more popular, simply because they are in categories that are more frequently visited by yelp users as a whole. Classification by category and zip code is crucial to comparing businesses fairly and getting the most out of the data.

The data used to determine classification must be from the tables outlined in Milestone 2.

A few tables were created based off of the updated ER diagram and json data set, a few key attributes from each will be used in the calculations behind business categorization. The Business table contains an address ID with the unique address (including the zipcode), which will be used to sort businesses. The table also includes a totalCheckins attribute, which calculates the total number of check-ins each day and hour for each business, which will be needed to determine popularity. The averageReviewRating attribute is calculated in a similar way and will be used for success metrics, with a higher average rating corresponding to higher success.

The exact calculation using the data is to be determined. However, the proposal for popularity is as follows: Take the sum of total reviews and total check-ins. Each business should be separated by zip code and category and those in the top 25% or the 75th percentile can be deemed “popular”. Popularity can also be made into some kind of score in between 0 and 1 rather than a binary, this will make it easier to determine success as our definition relies upon popularity and ratings.

For the comparison of all businesses in the same category, we may need to add a weight based on local population size to provide a fair comparison. Similarly, for the analysis of all business in the same zipcode, we may need to apply a weight to balance differences in the popularity of each business category.

For business success, we will take our popularity findings and add the context of review scores. The average score can be multiplied by the popularity score and a higher value will denote that a business has high customer volume and is highly rating by those who visit it. Like with popularity, businesses should be ranked by zip code or category and those in the highest percentile can be

deemed successful. Outlines for this milestone also suggested using business establishment date to determine success (under the guise that successful businesses stay longer due to consistent loyal customers), the jsons and er diagram provided thus far do not provide the data for this, but this can be a helpful metric in the future.

Lastly, findings may be put in a separate table to make user interaction easier. Tables for popularity may be divided by business zip code or business type/category, and will include business ID and popularity score, being ranked from top to bottom from most to least popular. The same plan will follow for storing and displaying findings for success. A feature to return user queries of popular or successful business may also be useful, simply returning a list of each business ID with a score in the high percentile range.